

Introduction to Functional Analysis

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1 Banach space theory

1.1 Quotient spaces, Baire category and uniform boundedness

Theorem. Let $\|\cdot\|$ be a **seminorm** on a vector space V . If we define $E = \{v \in V : \|v\| = 0\}$, then E is a subspace of V , and the function on V/E defined by

$$\|v + E\| = \|v\|$$

for any $v + E \in V/E$ defines a **norm**.

Theorem (Baire Category Theorem). Let M be a complete metric space, and let $\{C_n\}_{n=1}^\infty$ be a collection of closed subsets of M such that $M = \bigcup_{n=1}^\infty C_n$. Then at least one of the C_n contains an open ball $B(x, r) = \{y \in M : d(x, y) < r\}$.

Theorem (Uniform Boundedness Theorem). Let B be Banach space and V a normed vector space. Let $\{T_n\}_{n=1}^\infty$ be a sequence in $\mathcal{B}(B, V)$. Then if for all $b \in B$ we have $\sup_n \|T_n b\| < \infty$ (that is, this sequence is pointwise bounded), then $\sup_n \|T_n\| < \infty$ (the operator norms are bounded).

Proof. For each $k \in \mathbb{N}$, define

$$C_k = \left\{ b \in B : \|b\| \leq 1, \sup_{n \in \mathbb{N}} \|T_n b\| \leq k \right\}.$$

This set is closed for each $k \in \mathbb{N}$, but by assumption, we have

$$\{b \in B : \|b\| \leq 1\} = \bigcup_{k=1}^\infty C_k.$$

The left hand side is a closed subset of B , and is thus a complete metric space. By Baire Category Theorem, there exists $k \in \mathbb{N}$ such that C_k contains an open ball $B(b_0, \delta_0)$. Then, if $b \in B(0, \delta_0)$, we have $b_0 + b \in B(b_0, \delta_0)$ and thus

$$\sup_{n \in \mathbb{N}} \|T_n(b_0 + b)\| \leq k.$$

It follows that

$$\sup_{n \in \mathbb{N}} \|T_n b\| \leq \sup_{n \in \mathbb{N}} \|T_n(b_0 + b)\| + \sup_{n \in \mathbb{N}} \|T_n b_0\| \leq 2k.$$

Suppose $\|b\| = 1$, then $\frac{\delta_0}{2}b \in B(0, \delta_0)$ and thus for all $n \in \mathbb{N}$, we have

$$\left\| T_n \left(\frac{\delta_0}{2} b \right) \right\| \leq 2k.$$

Therefore,

$$\sup_{n \in \mathbb{N}} \|T_n\| \leq \frac{4k}{\delta_0}.$$

□

2 Hilbert space theory

2.1 Basic Hilbert space theory

Definition (Pre-Hilbert space). A **pre-Hilbert** space H is a vector space over $\mathbb{C}$ with a **Hermitian inner product**, which is a map $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}$ satisfying the following properties.

1. For all $\lambda_1, \lambda_2 \in \mathbb{C}$ and $v_1, v_2, w \in H$, we have

$$\langle \lambda_1 v_1 + \lambda_2 v_2, \lambda_1 v_1 + \lambda_2 v_2 \rangle = \lambda_1 \langle v_1, v_1 \rangle + \lambda_2 \langle v_2, v_2 \rangle.$$

2. For all $v, w \in H$, we have $\langle v, v \rangle = \overline{\langle w, w \rangle}$.
3. For all $v \in H$, we have $\langle v, v \rangle \geq 0$, with equality if and only if $v = 0$.

Definition. Let H be a pre-Hilbert space. For all $v \in H$, we define

$$\|v\| = \langle v, v \rangle^{\frac{1}{2}}.$$

Theorem (Cauchy-Schwarz inequality). Let H be a pre-Hilbert space. For all $u, v \in H$, we have

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Proof. Define $f(t) = \|u + tv\|^2$. Notice that

$$\begin{aligned} f(t) &= \langle u + tv, u + tv \rangle \\ &= \langle u, u \rangle + t^2 \langle v, v \rangle + t \langle u, v \rangle + t \langle v, u \rangle \\ &= \|u\|^2 + t^2 \|v\|^2 + 2t \operatorname{Re}(\langle u, v \rangle). \end{aligned}$$

This implies that

$$0 \leq f(t_{\min}) = \|u\|^2 - \frac{\operatorname{Re}(\langle u, v \rangle)^2}{\|v\|^2}.$$

It follows that

$$|\operatorname{Re}(\langle u, v \rangle)| \leq \|u\| \|v\|.$$

This is almost what we want. To finish up, first note that if $\langle u, v \rangle = 0$ then there is nothing to prove, so suppose $\langle u, v \rangle \neq 0$, and define

$$\lambda = \frac{\overline{\langle u, v \rangle}}{|\langle u, v \rangle|}.$$

Note that we have $|\lambda| = 1$ and we have the chain of equalities of real numbers:

$$|\langle u, v \rangle| = \lambda \langle u, v \rangle = \langle \lambda u, v \rangle = \operatorname{Re} \langle \lambda u, v \rangle \leq \|\lambda u\| \|v\|.$$

However, $\|\lambda u\| = \|u\|$, so the proof is complete. □

Theorem. If H is a pre-Hilbert space, then $\|\cdot\|$ is a norm on H .

Proof. Note that

$$\|v\| = 0 \iff \langle v, v \rangle = 0 \iff v = 0.$$

Now if $\lambda \in \mathbb{C}$ and $v \in H$, then

$$\langle \lambda v, \lambda v \rangle = \lambda \bar{\lambda} \langle v, v \rangle = |\lambda|^2 \|v\|^2.$$

Therefore, $\|\lambda v\| = |\lambda| \|v\|$.

Finally, let $u, v \in H$, then

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \|u\|^2 + \|v\|^2 + 2 \operatorname{Re} \langle u, v \rangle \\ &\leq \|u\|^2 + \|v\|^2 + 2 |\langle u, v \rangle| \\ &\leq \|u\|^2 + \|v\|^2 + 2 \|u\| \|v\| \\ &= (\|u\| + \|v\|)^2. \end{aligned}$$

This completes the proof. $\square$

Theorem. If $u_n \rightarrow u$ and $v_n \rightarrow v$ in a pre-Hilbert space H , then $\langle u_n, v_n \rangle \rightarrow \langle u, v \rangle$.

Proof. If $u_n \rightarrow u$ and $v_n \rightarrow v$, then $\|u_n - u\| \rightarrow 0$ and $\|v_n - v\| \rightarrow 0$. It follows that

$$\begin{aligned} |\langle u_n, v_n \rangle - \langle u, v \rangle| &= |\langle u_n - u, v_n - v \rangle + \langle u, v_n - v \rangle| \\ &\leq |\langle u_n - u, v_n - v \rangle| + |\langle u, v_n - v \rangle| \\ &\leq \|u_n - u\| \|v_n - v\| + \|u\| \|v_n - v\| \\ &\leq \|u_n - u\| \sup_{k \in \mathbb{N}} \|v_k\| + \|u\| \|v_n - v\| \\ &\rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. This completes the proof. $\square$

Definition (Hilbert space). A **Hilbert space** is a pre-Hilbert space that is complete with respect to the norm $\|\cdot\| = \langle \cdot, \cdot \rangle^{\frac{1}{2}}$.

Example. Some examples of Hilbert spaces:

- $\mathbb{C}^n = \{z = (z_1, \dots, z_n) : z_j \in \mathbb{C}\}$ with $\langle z, z \rangle = \sum_j z_j \bar{z}_j$ is a Hilbert space.
- $\ell^2 = \left\{a = \{a_k\}_{k=1}^\infty : a_k \in \mathbb{C}, \sum_{k=1}^\infty |a_k|^2 < \infty\right\}$ with $\langle a, a \rangle = \sum_{k=1}^\infty a_k \bar{a}_k$ is a Hilbert space.
- If $E \subset \mathbb{R}$ is measurable, then $L^2(E) = \left\{f : E \rightarrow \mathbb{C}, \int_E |f|^2 < \infty\right\}$ with $\langle f, f \rangle = \int_E f \bar{g}$ is a Hilbert space.

We will show that each separable Hilbert space is isometrically isomorphic to either $\mathbb{C}^n$ or ℓ^2 .

Now we have seen that ℓ^2 and L^2 spaces are Hilbert spaces. This is expected since the definition of the inner product in these spaces uses the fact that they are ℓ^2 or L^2 . A natural question then is whether other ℓ^p or L^p spaces are also Hilbert spaces with respect to some inner product? It turns out there is a simple way to decide whether a norm come from an inner-product, and thus whether a Banach space is a Hilbert space.

Theorem (Parallelogram Law). If H is a pre-Hilbert space, then for all $u, v \in H$, we have

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2).$$

In addition, if H is a normed vector space satisfying this equality, then H is a pre-Hilbert space.

Using the previous theorem, we can verify that ℓ^p and L^p with $p \neq 2$ are **not** Hilbert spaces.

Definition (Orthogonal). If H is a pre-Hilbert space, $u, v \in H$ are **orthogonal** if $\langle u, v \rangle = 0$. We denote this as $u \perp v$.

Definition (Orthonormal sets). If H is a pre-Hilbert space, a subset $\{e_\lambda\}_{\lambda \in \Lambda} \subset H$ is **orthonormal** if for all $\lambda \in \Lambda$, we have $\|e_\lambda\| = 1$ and $\lambda_1 \neq \lambda_2$ implies $e_{\lambda_1} \perp e_{\lambda_2}$.

Remark. we will mainly be interested in the case where we have a countable orthonormal set.

Example. The set $\left\{\frac{e^{inx}}{\sqrt{2\pi}}\right\}_{n \in \mathbb{Z}}$ as elements in $L^2([-\pi, \pi])$ is an orthonormal subset of $L^2([-\pi, \pi])$. Indeed, for any $m, n \in \mathbb{Z}$, we have

$$\int_{-\pi}^{\pi} e^{imx} \overline{e^{inx}} = \int_{-\pi}^{\pi} e^{i(m-n)x} = \begin{cases} 2\pi & (m = n), \\ 0 & (m \neq n). \end{cases}$$

Therefore, $\left\langle \frac{e^{inx}}{\sqrt{2\pi}}, \frac{e^{inx}}{\sqrt{2\pi}} \right\rangle = \delta_{mn}$, and $\left\{\frac{e^{inx}}{\sqrt{2\pi}}\right\}_{n \in \mathbb{Z}}$ is an orthonormal subset of $L^2([-\pi, \pi])$.

Theorem (Bessel). If $\{e_n\}_{n=1}^\infty$ is countable orthonormal subset of a pre-Hilbert space H , then for all $u \in H$, we have

$$\sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2 \leq \|u\|^2.$$

Proof. We first do the finite case. Suppose $\{e_n\}_{n=1}^N$ is an orthonormal subset of H . Then,

$$\begin{aligned} \left\| \sum_{n=1}^N \langle u, e_n \rangle e_n \right\|^2 &= \left\langle \sum_{n=1}^N \langle u, e_n \rangle e_n, \sum_{n=1}^N \langle u, e_n \rangle e_n \right\rangle \\ &= \sum_{n=1}^N \sum_{m=1}^N \langle u, e_n \rangle \overline{\langle u, e_m \rangle} \langle e_n, e_m \rangle \\ &= \sum_{n=1}^N |\langle u, e_n \rangle|^2. \end{aligned}$$

Also,

$$\begin{aligned}\langle u, u \rangle &= \sum_{n=0}^N \overline{\langle u, u \rangle} \langle u, u \rangle \\ &= \sum_{n=0}^N |\langle u, u \rangle|^2.\end{aligned}$$

Therefore,

$$\begin{aligned}0 &\leq \left\| u - \sum_{n=0}^N \langle u, u \rangle e_n \right\|^2 \\ &= \|u\|^2 + \left\| \sum_{n=0}^N \langle u, u \rangle e_n \right\|^2 - 2 \operatorname{Re} \langle u, u \rangle \\ &= \|u\|^2 - \sum_{n=0}^N |\langle u, u \rangle|^2,\end{aligned}$$

as desired.

For the infinite case, just take the limit as $N \rightarrow \infty$. $\square$

Definition (Maximal orthonormal subset). An orthonormal subset $\{e_\lambda\}_\lambda$ of a pre-Hilbert space is **maximal** if $u \in H$ and $\langle u, u \rangle = 0$ for all $\lambda \in \Lambda$ implies that $u = 0$.

Theorem. Every non-trivial pre-Hilbert space has a maximal orthonormal subset.

This can be proved using Zorn's Lemma. We will prove something less strong but often equally useful by hand, without applying Zorn's Lemma.

Theorem. Every non-trivial separable pre-Hilbert space has a countable maximal orthonormal subset.

Proof. Use the Gram-Schmidt process. Let $\{v_j\}_{j=1}^\infty$ be a countable dense subset of H where $v_0 \neq 0$. Claim that for any $n \in \mathbb{N}$, there exists $m(n) \leq n$ and an orthonormal subset $\{e_1, \dots, e_{m(n)}\}$ such that

1. $\operatorname{span} \{e_1, \dots, e_{m(n)}\} = \operatorname{span} \{v_1, \dots, v_n\}$.
2. If $v_n \in \operatorname{span} \{v_1, \dots, v_{n-1}\}$, we have

$$\{e_1, \dots, e_{m(n)}\} = \{e_1, \dots, e_{m(n-1)}\} \cup \emptyset.$$

Otherwise, we have

$$\{e_1, \dots, e_{m(n)}\} = \{e_1, \dots, e_{m(n-1)}\} \cup e_{m(n)}$$

for some $e_{m(n)} \in H$.

Prove this by induction. For the base case, let $e_1 = \frac{v_1}{\|v_1\|}$. For the inductive step, suppose the claim holds for $n = k$. If $v_{k+1} \in \text{span}\{v_1, \dots, v_k\}$, then

$$\text{span}\{e_1, \dots, e_{n(k)}\} = \text{span}\{v_1, \dots, v_k\} = \text{span}\{v_1, \dots, v_{k+1}\}.$$

Now suppose $v_{k+1} \notin \text{span}\{v_1, \dots, v_k\}$. Define

$$w_{k+1} = v_{k+1} - \sum_{j=1}^{m(k)} \langle v_{k+1}, v_{k+1} \rangle e_j.$$

Note that $w_{k+1} \neq 0$ and define $e_{m(k+1)} = \frac{w_{k+1}}{\|w_{k+1}\|}$. Then, $\|e_{m(k+1)}\| = 1$ and for all $1 \leq l \leq m(k)$,

$$\begin{aligned} \langle e_{m(k+1)}, e_{m(k+1)} \rangle &= \frac{1}{\|w_{k+1}\|} \left\langle v_{k+1} - \sum_{j=1}^{m(k)} \langle v_{k+1}, v_{k+1} \rangle e_j, v_{k+1} - \sum_{j=1}^{m(k)} \langle v_{k+1}, v_{k+1} \rangle e_j \right\rangle \\ &= \frac{1}{\|w_{k+1}\|} (\langle v_{k+1}, v_{k+1} \rangle - \langle v_{k+1}, v_{k+1} \rangle) \\ &= 0. \end{aligned}$$

Therefore, $e_{m(k+1)}$ is the desired vector we want and we have completed the proof for the claim.

Now let

$$S = \bigcup_{n=1}^{\infty} \{e_1, \dots, e_{m(n)}\}.$$

Then S is a countable orthonormal subset of H . Now we show S is maximal. Suppose $u \in H$ and $\langle u, u \rangle = 0$. Since $\{v_j\}_{j=1}^{\infty}$ is dense in H , there exists $\{v_{j(k)}\}_{k=1}^{\infty}$ such that $v_{j(k)} \rightarrow u$ as $k \rightarrow \infty$. By our claim, we know $v_{j(k)} \in \text{span}\{e_1, \dots, e_{m(j(k))}\}$. By Bessel's inequality,

$$\|v_{j(k)}\|^2 = \sum_{l=1}^{m(j(k))} |\langle v_{j(k)}, v_{j(k)} \rangle|^2 = \sum_{l=1}^{m(j(k))} |\langle v_{j(k)} - u, v_{j(k)} - u \rangle|^2 \leq \|v_{j(k)} - u\|^2,$$

where for the first equality we used the fact that $v_{j(k)} \in \text{span}\{e_1, \dots, e_{m(j(k))}\}$. Since $v_{j(k)} \rightarrow u$ as $k \rightarrow \infty$, the inequality implies that $\|v_{j(k)}\| \rightarrow 0$ as $k \rightarrow \infty$ and thus $\|u\| = 0$, showing that S is indeed a maximal orthonormal subset of H . $\square$

Corollary. ℓ^2 and L^2 have countable maximal orthonormal subset since they are both separable.

2.2 Orthonormal bases and Fourier Series

Definition (Orthonormal basis). Let H be a Hilbert space. An **orthonormal basis** of H is a countable maximal orthonormal subset $\{e_n\}_n$ of H .

Theorem. If $\{e_n\}_{n=1}^{\infty}$ is an orthonormal basis in Hilbert space H , then for all $u \in H$, we have

$$\sum_{n=1}^{\infty} \langle u, u \rangle e_n = u.$$

This is the Fourier-Bessel series.

This tells us we can write each element in H as a infinite linear combination of the orthonormal basis.

Proof. We first prove the sequence of partial sums $\{\sum_{n=0}^m \langle u, u \rangle e_n\}_m$ is Cauchy. Let $\varepsilon > 0$. By Bessel's inequality, we have

$$\sum_{n=1}^{\infty} |\langle u, u \rangle|^2 \leq \|u\|^2 < \infty.$$

Therefore, there exists $M \in \mathbb{N}$ such that $N \geq M$ implies $\sum_{n=N+1}^{\infty} |\langle u, u \rangle|^2 < \varepsilon^2$. Then for all $m > l \geq M$, we have

$$\left\| \sum_{n=0}^m \langle u, u \rangle e_n - \sum_{n=0}^l \langle u, u \rangle e_n \right\|^2 \leq \sum_{n=l+1}^m |\langle u, u \rangle|^2 \leq \sum_{n=l+1}^{\infty} |\langle u, u \rangle|^2 < \varepsilon^2.$$

Therefore, the sequence of partial sum is Cauchy. Since H is complete, there exists $\bar{u} \in H$ such that $\bar{u} = \sum_{n=1}^{\infty} \langle u, u \rangle e_n$. It remains to show that $\bar{u} = u$. By continuity of inner-product, for all $l \in \mathbb{N}$, we have

$$\begin{aligned} \langle u - \bar{u}, u - \bar{u} \rangle &= \lim_{m \rightarrow \infty} \left\langle u - \sum_{n=0}^m \langle u, u \rangle e_n, u - \sum_{n=0}^m \langle u, u \rangle e_n \right\rangle \\ &= \lim_{m \rightarrow \infty} \left[\langle u, u \rangle - \sum_{n=0}^m \langle u, u \rangle \langle e_n, e_n \rangle \right] \\ &= 0. \end{aligned}$$

Since $\{e_n\}_{n=1}^{\infty}$ is maximal, this implies that $u - \bar{u} = 0$ and the proof is complete. $\square$

Theorem. Let H be a Hilbert space. If H has an orthonormal basis, then H is separable.

Proof. Suppose $\{e_n\}_{n=1}^{\infty}$ is an orthonormal basis for H . Then

$$S = \bigcup_{m \in \mathbb{N}} \left\{ \sum_{n=0}^m q_n e_n : q_n \in \mathbb{Q} + i\mathbb{Q} \right\}$$

is a countable set. Also, by the previous theorem, S is dense in H . $\square$

Remark. Let H be a Hilbert space. H is separable if and only if H has an orthonormal basis.

Theorem (Parseval's identity). If H is a Hilbert space and $\{e_n\}_{n=1}^{\infty}$ is a countable orthonormal basis, then for all $u \in H$, we have

$$\sum_n |\langle u, e_n \rangle|^2 = \|u\|^2$$

Proof. We have $u = \sum_n \langle u, e_n \rangle e_n$. This implies that

$$\begin{aligned} \|u\|^2 &= \lim_{m \rightarrow \infty} \left\langle \sum_{n=0}^m \langle u, e_n \rangle e_n, \sum_{n=0}^m \langle u, e_n \rangle e_n \right\rangle \\ &= \lim_{m \rightarrow \infty} \sum_{n=0}^m \sum_{l=0}^m \langle u, e_n \rangle \overline{\langle u, e_l \rangle} \langle e_n, e_l \rangle \\ &= \lim_{m \rightarrow \infty} \sum_{n=0}^m |\langle u, e_n \rangle|^2 \\ &= \sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2. \end{aligned}$$

□

Theorem. If H is an infinite dimensional separable Hilbert space, then H is isometrically isomorphic to ℓ^2 . That is, there exists bijective bounded linear map $T : H \rightarrow \ell^2$ such that for all $u, v \in H$, we have

$$\|Tu\|_{\ell^2} = \|u\|_H \quad \text{and} \quad \langle Tu, Tv \rangle_{\ell^2} = \langle u, v \rangle_H.$$

Proof. Since H is separable, there exists an orthonormal basis $\{e_n\}_{n=1}^{\infty}$. For all $u \in H$, the previous theorem gives

$$\|u\| = \left(\sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2 \right)^{\frac{1}{2}}.$$

Define $T : H \rightarrow \ell^2$ by

$$Tu = \{\langle u, e_n \rangle\}_{n=1}^{\infty} \in \ell^2.$$

It is easy to check that T is the desired isometric isomorphism. □

Next we use the theories we learned in a more concrete setting — the Fourier series.

Theorem. The subset $\left\{ \frac{e^{inx}}{\sqrt{2\pi}} \right\}_{n \in \mathbb{Z}}$ is an orthonormal subset of $L^2([-\pi, \pi])$.

Definition. Let $f \in L^2([-\pi, \pi])$. Then the n -th **Fourier coefficient** of f is

$$\widehat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt.$$

The N -th Fourier sum of f is

$$S_N f(x) = \sum_{|n| \leq N} \widehat{f}(n) e^{inx} = \sum_{|n| \leq N} \langle f, f \rangle \frac{e^{inx}}{\sqrt{2\pi}}.$$

The **Fourier series** of f is the formal series $\sum_{n \in \mathbb{Z}} \widehat{f}(n) e^{-inx}$.

The natural question now is whether we have for all $f \in L^2([-\pi, \pi])$,

$$f(x) = \sum_{n \in \mathbb{Z}} \widehat{f}(n) e^{inx}.$$

That is, whether we have the following convergence in L^2 .

$$\lim_{N \rightarrow \infty} \|f - S_N f\|_2 = 0.$$

This question is then equivalent to whether $\left\{ \frac{e^{inx}}{\sqrt{2\pi}} \right\}_{n \in \mathbb{Z}}$ is maximal in $L^2([-\pi, \pi])$. That is, whether $\widehat{f}(n) = 0$ for all $n \in \mathbb{N}$ implies $f = 0$.

The answer to the question is yes, but it is going to take some work. We first do some simple calculation.

Theorem. For all $f \in L^2([-\pi, \pi])$ and for all $N \in \mathbb{N}$, we have

$$S_N f(x) = \int_{-\pi}^{\pi} D_N(x-t) f(t) dt,$$

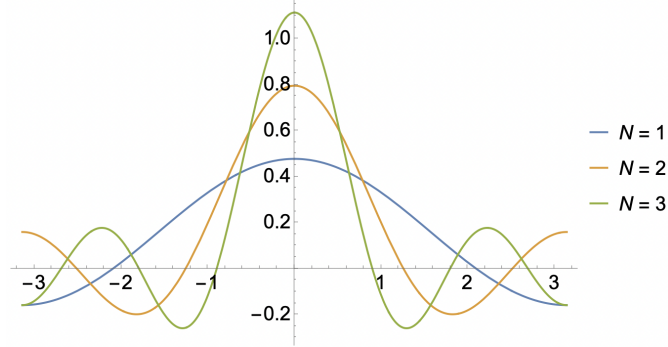
where

$$D_N(x) = \begin{cases} \frac{2N+1}{2\pi} & (x = 0) \\ \frac{\sin(N+\frac{1}{2})x}{2\pi \sin \frac{x}{2}} & (x \neq 0) \end{cases}$$

it the **Dirichlet kernel**. Figure 1 shows a plot of $D_N(x)$ on $[-\pi, \pi]$ for $N = 1, 2, 3$. Note that D_N is a smooth function.

Proof. If $f \in L^2([-\pi, \pi])$, we have

$$\begin{aligned} S_N f(x) &= \sum_{|n| \leq N} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt \right) e^{inx} \\ &= \int_{-\pi}^{\pi} f(t) \left(\frac{1}{2\pi} \sum_{|n| \leq N} e^{in(x-t)} \right) dt. \end{aligned}$$

Figure 1: Plot of Dirichlet kernel $D_N(x)$ on $[-\pi, \pi]$ for $N = 1, 2, 3$.

Let $D_N(x) = \frac{1}{2\pi} \sum_{|n| \leq N} e^{in(x-t)}$. Then for $x \neq 0$, we have

$$\begin{aligned}
 D_N(x) &= \frac{1}{2\pi} \sum_{n=-N}^N e^{-inx} \\
 &= \frac{1}{2\pi} e^{-iNx} \sum_{n=0}^{2N} (e^{ix})^n \\
 &= \frac{1}{2\pi} e^{-iNx} \frac{1 - e^{i(2N+1)x}}{1 - e^{ix}} \\
 &= \frac{1}{2\pi} \frac{e^{i(N+\frac{1}{2})x} - e^{-i(N+\frac{1}{2})x}}{e^{\frac{ix}{2}} - e^{-\frac{ix}{2}}} \\
 &= \frac{1}{2\pi} \frac{2i \sin(N + \frac{1}{2})x}{2i \sin \frac{x}{2}} \\
 &= \frac{1}{2\pi} \frac{\sin(N + \frac{1}{2})x}{\sin \frac{x}{2}},
 \end{aligned}$$

as desired. For $x = 0$, we also clearly have $D_N(0) = \frac{(2N+1)}{2\pi}$. The proof is thus complete. $\square$

Definition. If $f \in L^2([-\pi, \pi])$, we define the N -th **Cesaro-Fourier mean** of f by

$$\sigma_N f(x) = \frac{1}{N+1} \sum_{k=0}^N S_k f(x).$$

The idea behind defining the Cesaro mean is that if the original sequence converges, the Cesaro mean also converge to the same limit. However, Cesaro have even better property — the Cesaro mean can converge even if the original sequence does not converge. Therefore, it has better

convergence properties and hopefully we can show it converge to f in L^2 more easily. The goal now is then to show

$$\|\sigma_N f - f\|_2 \rightarrow 0 \text{ as } N \rightarrow \infty.$$

This would tell us if all Fourier coefficients are zero, then the Cesaro means are zero, and the limit above would tell us f is zero.

2.3 Fejer's theorem and convergence of Fourier series

In this section, we will show that if $f \in L^2([-\pi, \pi])$, then $\|\sigma_N f - f\|_2 \rightarrow 0$ as $N \rightarrow \infty$.

First we will rewrite the Cesaro Fourier mean, just like what we did for the partial Fourier sum using the Dirichlet kernel.

Theorem. For any $f \in L^2([-\pi, \pi])$, we have

$$\sigma_N f(x) = \int_{-\pi}^{\pi} K_N(x-t) f(t) dt,$$

where

$$K_N(x) = \begin{cases} \frac{N+1}{2\pi} & (x = 0) \\ \frac{1}{2\pi(N+1)} \frac{\sin^2 \frac{N+1}{2} x}{\sin^2 \frac{x}{2}} & (x \neq 0) \end{cases}$$

is the Fejer kernel.

Moreover, we have

1. $K_N(x) \geq 0$, $K_N(x) = K_N(-x)$, and $K_N(x)$ is 2π periodic.
2. $\int_{-\pi}^{\pi} K_N(t) dt = 1$.
3. If $\delta \in (0, \pi]$, then for all $\delta \leq |x| \leq \pi$, we have

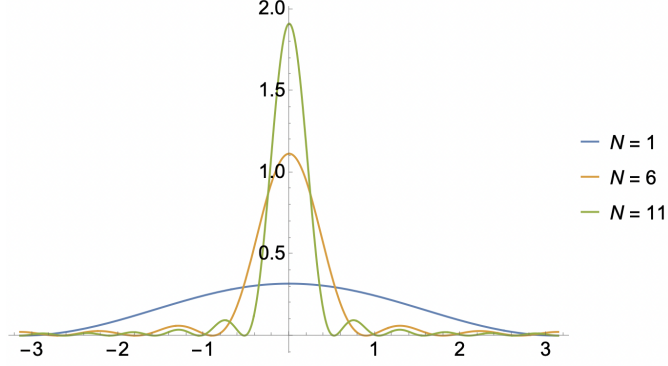
$$|K_N(x)| \leq \frac{1}{2\pi(N+1) \sin^2 \frac{\delta}{2}}.$$

A plot for $K_N(x)$ is shown in Figure 2.

Note that $K_N(x)$ is concentrated at 0 when N is very large. In this case, we have

$$\begin{aligned} \sigma_N f(x) &= \int_{-\pi}^{\pi} K_N(x-t) f(t) dt \\ &\approx f(x) \int_{-\pi}^{\pi} K_N(t) dt \\ &= f(x). \end{aligned}$$

This provides a rough intuition behind the Fejer kernel. The fact that K_N is non-negative makes a huge difference compared to the Dirichlet kernel, since it gives much better properties.

Figure 2: Plot of Dirichlet kernel $D_N(x)$ on $[-\pi, \pi]$ for $N = 1, 6, 11$.

Proof. Recall that

$$S_k f(x) = \int_{-\pi}^{\pi} D_k(x-t) f(t) dt,$$

where

$$D_k(t) = \begin{cases} \frac{2N+1}{2\pi} & (t = 0), \\ \frac{1}{2\pi} \frac{\sin(N+\frac{1}{2})t}{\sin \frac{t}{2}} & (t \neq 0). \end{cases}$$

It follows that

$$\begin{aligned} \sigma_N f(x) &= \frac{1}{N+1} \sum_{k=0}^N S_k f(x) \\ &= \int_{-\pi}^{\pi} \frac{1}{N+1} \sum_{k=0}^N D_k(x-t) f(t) dt. \end{aligned}$$

Then for $x \neq 0$, we have

$$\begin{aligned} K_N(x) &= \frac{1}{N+1} \sum_{k=0}^N D_k(x) \\ &= \frac{1}{2\pi(N+1)} \frac{1}{2\sin^2 \frac{x}{2}} \sum_{k=0}^N 2\sin \frac{x}{2} \sin \left(k + \frac{1}{2}\right) x \\ &= \frac{1}{2\pi(N+1)} \frac{1}{2\sin^2 \frac{x}{2}} \sum_{k=0}^N [\cos kx - \cos(k+1)x] \\ &= \frac{1}{2\pi(N+1)} \frac{1}{\sin^2 \frac{x}{2}} \frac{1 - \cos(N+1)x}{2} \\ &= \frac{1}{2\pi(N+1)} \frac{\sin^2 \frac{N+1}{2} x}{\sin^2 \frac{x}{2}}. \end{aligned}$$

It follows immediately that $K_N(x) \geq 0$, $K_N(x)$ is even and 2π periodic.

For property 2, note that for all k ,

$$\int_{-\pi}^{\pi} D_k(t) dt = \int_{-\pi}^{\pi} \sum_{n=-k}^k e^{int} dt = 1.$$

Then,

$$\int_{-\pi}^{\pi} K_N(t) dt = \frac{1}{N+1} \sum_{k=0}^N \int_{-\pi}^{\pi} D_k(t) dt = 1,$$

as desired.

For property 3, let $\delta \in (0, \pi]$. Note that $\sin^2 \frac{x}{2}$ is even and increasing on $[0, \pi]$. It follows that $\delta \leq |x| \leq \pi$ implies $\sin^2 \frac{x}{2} \geq \sin^2 \frac{\delta}{2}$. Therefore,

$$K_N(x) \leq \frac{1}{2\pi(N+1)} \frac{\sin^2 \frac{N+1}{2}x}{\sin^2 \frac{\delta}{2}} \leq \frac{1}{2\pi(N+1) \sin^2 \frac{\delta}{2}}.$$

□

Since the continuous functions that vanishes at both end points is dense in $L^2([-\pi, \pi])$, it make sense to first prove the theorem for continuous functions. We have the following theorem by Fejer.

Theorem (Fejer). If $f \in C([-\pi, \pi])$ is 2π -periodic, $f(\pi) = f(-\pi)$, then $\sigma_N f \rightarrow f$ uniformly on $[-\pi, \pi]$.

Proof. First we extend f by periodicity to all of $\mathbb{R}$. Then $f \in C(\mathbb{R})$, 2π -periodic. This implies that f is uniformly continuous and bounded.

Let $\varepsilon > 0$. Since f is uniformly continuous, there exists $\delta > 0$ such that $|y - z| < \delta$ implies that $|f(y) - f(z)| < \frac{\varepsilon}{2}$. Choose $M \in \mathbb{N}$ such that

$$\frac{2 \|f\|_{\infty}}{(N+1) \sin^2 \frac{\delta}{2}} < \frac{\varepsilon}{2}.$$

for all $N \geq M$. Also, since f and K_N are both 2π -periodic, we have

$$\sigma_N f(x) = \int_{-\pi}^{\pi} K_N(x-t) f(t) dt = \int_{x-\pi}^{x+\pi} K_N(\tau) f(x-\tau) d\tau = \int_{-\pi}^{\pi} K_N(\tau) f(x-\tau) d\tau.$$

Then for all $N \geq M$ and for all $x \in [-\pi, \pi]$, we have

$$\begin{aligned}
|\sigma_N f(x) - f(x)| &= \left| \int_{-\pi}^{\pi} K_N(t) f(x-t) dt - \int_{-\pi}^{\pi} K_N(t) f(x) dt \right| \\
&\leq \int_{-\pi}^{\pi} K_N(t) |f(x-t) - f(x)| dt \\
&\leq \int_{|t| < \delta} K_N(t) |f(x-t) - f(x)| dt + \int_{\delta \leq |t| \leq \pi} K_N(t) |f(x-t) - f(x)| dt \\
&\leq \frac{\varepsilon}{2} \int_{|t| < \delta} K_N(t) dt + 2 \|f\|_{\infty} \int_{\delta \leq |t| \leq \pi} \frac{1}{2\pi(N+1) \sin^2 \frac{\delta}{2}} dt \\
&\leq \frac{\varepsilon}{2} + \frac{2 \|f\|_{\infty}}{(N+1) \sin^2 \frac{\delta}{2}} \\
&\leq \varepsilon.
\end{aligned}$$

□

Remark. The same proof can be modified if instead of $K_N(x) \geq 0$, we have

$$\sup_{N \in \mathbb{N}} \int_{-\pi}^{\pi} |K_N(x)| dx < \infty.$$

Note that

$$\int_{-\pi}^{\pi} |D_N(x)| dx \sim \log N,$$

so we cannot reproduct the proof using Dirichlet kernel.

We only need some last bit of information to conclude the answer of our main question.

Theorem. For all $f \in L^2([-\pi, \pi])$, we have $\|\sigma_N f\|_2 \leq \|f\|_2$.

Proof. Suppose first the $f \in C([-\pi, \pi])$ and 2π -periodic. Then $\sigma_N f(x) = \int_{-\pi}^{\pi} K_N(t) f(x-t) dt$. It follows that

$$\begin{aligned}
\int_{-\pi}^{\pi} |\sigma_N f(x)|^2 dx &= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} f(x-s) \overline{f(x-t)} K_N(s) K_N(t) ds dt dx \\
&= \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} K_N(s) K_N(t) \left[\int_{-\pi}^{\pi} f(x-s) \overline{f(x-t)} dx \right] ds dt \\
&\leq \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} K_N(s) K_N(t) \|f(\cdot - s)\|_2 \|f(\cdot - t)\|_2 ds dt \\
&\leq \|f\|_2^2 \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} K_N(s) K_N(t) ds dt \\
&= \|f\|_2^2,
\end{aligned}$$

where we used Cauchy-Schwarz inequality. This implies that $\|\sigma_N f\|_2 \leq \|f\|_2$.

Now for the general case, by density there exists sequence $\{f_n\}_{n=1}^\infty$ of 2π -periodic continuous function that $\|f_n - f\|_2 \rightarrow 0$. Then, $\|\sigma_N f_n - \sigma_N f\| \rightarrow 0$ as $n \rightarrow \infty$. Therefore,

$$\|\sigma_N f\|_2 = \lim_{n \rightarrow \infty} \|\sigma_N f_n\|_2 \leq \lim_{n \rightarrow \infty} \|f_n\|_2 = \|f\|_2.$$

□

Theorem. For all $f \in L^2([-\pi, \pi])$, we have $\|\sigma_N f - f\|_2 \rightarrow 0$ as $N \rightarrow \infty$. In particular, as a immediate corollary, if $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$.

Proof. Let $f \in L^2([-\pi, \pi])$ and $\varepsilon > 0$. Again by density there exists $g \in C([-\pi, \pi])$ 2π -periodic such that $\|f - g\|_2 \leq \frac{\varepsilon}{3}$. Since $\sigma_N g \rightarrow g$ uniformly on $[-\pi, \pi]$, there exists $M \in \mathbb{N}$ such that for all $N \geq M$ and all $x \in [-\pi, \pi]$, we have

$$|\sigma_N g(x) - g(x)| < \frac{\varepsilon}{3\sqrt{2\pi}}.$$

Then for all $N \geq M$,

$$\begin{aligned} \|\sigma_N f - f\|_2 &\leq \|\sigma_N(f - g)\|_2 + \|\sigma_N g - g\| + \|g - f\|_2 \\ &\leq 2\|f - g\|_2 + \left(\int_{-\pi}^{\pi} |\sigma_N g(x) - g(x)|^2 dx \right)^{\frac{1}{2}} \\ &\leq \varepsilon. \end{aligned}$$

□

Remark. We have shown that for all $f \in L^2([-\pi, \pi])$, $\|S_N f - f\|_2 \rightarrow 0$. This does not say $S_N f$ converge to f almost everywhere. However, by a theorem by Carleson, for all $f \in L^2([-\pi, \pi])$, we actually do have $S_N f \rightarrow f$ almost everywhere. Also, for all $1 < p < \infty$, $\|S_N f - f\|_p \rightarrow 0$. This is not true for $p = 1$ or $p = \infty$.

2.4 Minimizers, orthogonal complements, and Riesz representation theorem

Length minimizers

Theorem. Suppose H a Hilbert space and $C \subset H$ is a subset such that

1. $C \neq \emptyset$.
2. C is closed.
3. C is convex. That is, if $v_1, v_2 \in C$ and $t \in [0, 1]$, then $tv_1 + (1 - t)v_2 \in C$.

Then, there exists a unique $v \in C$ such that $\|v\| = \inf_{u \in C} \|u\|$.

Proof. Let $d = \inf_{u \in C} \|u\|$, which we know exists. Then there exists sequence $\{u_n\}_{n=1}^\infty \subset C$ such that $\|u_n\| \rightarrow d$.

Claim that $\{u_n\}_{n=1}^\infty$ is Cauchy. Let $\varepsilon > 0$ be given. There exists $N \in \mathbb{N}$ such that $n \geq N$ implies

$$2\|u_n\|^2 < 2d^2 + \frac{\varepsilon^2}{2}.$$

It follows that for all $n, m \geq N$, we have

$$\|u_n - u_m\|^2 = 2\|u_n\|^2 + 2\|u_m\|^2 - 4\left\|\frac{u_n + u_m}{2}\right\|^2,$$

by the Parallelogram law. Note that $\frac{u_n + u_m}{2} \in C$. Therefore,

$$\|u_n - u_m\|^2 \leq 2d^2 + \frac{\varepsilon^2}{2} + 2d^2 + \frac{\varepsilon^2}{2} - 4d^2 = \varepsilon^2.$$

This shows that $\{u_n\}_{n=1}^\infty$ is Cauchy.

Since H is Hilbert space, there exists $v \in H$ such that $u_n \rightarrow v$. Since C is closed, $v \in C$. It is also clear that $\|v\| = d$. To show this element is unique, suppose $v, \bar{v} \in C$ and $\|v\| = \|\bar{v}\| = d$. Then,

$$\|v - \bar{v}\|^2 = 2\|v\|^2 + 2\|\bar{v}\|^2 - 4\left\|\frac{v + \bar{v}}{2}\right\|^2 \leq 4d^2 - 4d^2 = 0.$$

This implies that $v = \bar{v}$ and the proof is complete. $\square$

Orthocomplements

Theorem. If H is a Hilbert space, $W \subset H$ is a subspace, then

$$W^\perp = \{u \in H : \langle u, w \rangle = 0 \text{ for all } w \in W\}$$

is a closed linear subspace of H .

Moreover, if W is closed, then

$$H = W \oplus W^\perp.$$

That is, for all $u \in H$, there exists unique $w \in W$ and $w^\perp \in W^\perp$ such that $u = w + w^\perp$.

Proof. It is easy to show that $W^\perp$ is a subspace of H , and $W \cap W^\perp = \{0\}$. To show $W^\perp$ is closed, let $\{u_n\}_{n=1}^\infty$ be a sequence in $W^\perp$ and $u \in H$ such that $u_n \rightarrow u$. We need to show that $u \in W^\perp$. Let $w \in W$, then

$$\langle u, w \rangle = \lim_{n \rightarrow \infty} \langle u_n, w \rangle = 0.$$

Therefore, $u \in W^\perp$ and $W^\perp$ is a closed linear subspace of H .

Now suppose W is closed. If $W = H$, then $W^\perp = \{0\}$ and $H = W \oplus W^\perp$. Now assume that $W \neq H$. Let $u \in H \setminus W$ and define

$$C = u + W = \{u + w : w \in W\}.$$

Note that $u \in C$ so $C \neq \emptyset$. Also, C is convex, since if $u + w_1 \in C$, $u + w_2 \in C$, and $t \in [0, 1]$, then

$$t(u + w_1) + (1 - t)(u + w_2) = u + (tw_1 + (1 - t)w_2) \in u + W.$$

Claim that C is also closed. Suppose $\{u + w_n\}_{n=1}^\infty \subset C$ is such that $u + w_n \rightarrow v$ for some $v \in H$. We want to show that $v \in C$. This implies that $w_n \rightarrow v - u$ and since W is closed, $v - u \in W$. It follows that $v = u + (v - u)$ so $v \in C$.

Since C is nonempty, closed, and convex, there exists unique element $v \in C$ such that

$$\|v\| = \inf_{w \in W} \|u + w\|.$$

Note that $v \in C$ so $u - v \in W$. Also, $u = (u - v) + v$. Claim that $v \in W^\perp$. Let $w \in W$ and

$$f(t) = \|v + tw\|^2 = \|v\|^2 + t^2 \|w\|^2 + 2t \operatorname{Re} \langle v, w \rangle.$$

Then $f(t)$ has a minimum at $t = 0$, which implies $f'(0) = \langle v, w \rangle = 0$. Repeat the previous argument with iw in place of w to obtain $\operatorname{Re} \langle v, w \rangle = \operatorname{Im} \langle v, w \rangle = 0$. This shows that $w \in W^\perp$ and thus $H = W + W^\perp$.

To show the decomposition is unique, suppose $u = w_1 + w_1^\perp = w_2 + w_2^\perp$. This implies that

$$w_2 - w_1 = w_1^\perp - w_2^\perp \in W \cap W^\perp.$$

However, $W \cap W^\perp = \{0\}$, so $w_1 = w_2$ and $w_1^\perp = w_2^\perp$. □

Theorem. If $W \subset H$ is a subspace, then

$$\overline{W} = (W^\perp)^\perp,$$

where $\overline{W}$ is the closure of W .

Proof. Homework. □

Definition (Projection). A bounded linear operator $P : H \rightarrow H$ is a **projection** if $P^2 = P$.

Theorem. Let H be a Hilbert space, $W \subset H$ be a closed subspace. Then by the previous theorem we have $H = W \oplus W^\perp$. Define $\Pi_W : H \rightarrow H$ in the following way: for $v = w + w^\perp$, define

$$\Pi_W(v) = w.$$

Then Π_W is a projection.

Proof. It is easy to verify that Π_W is linear and $\Pi_W^2 = \Pi_W$. Claim Π_W is bounded. Suppose $v = w + w^\perp$. It follows that

$$\|v\|^2 = \|w + w^\perp\|^2 = \|w\|^2 + \|w^\perp\|^2 \geq \|w\|^2.$$

This shows that $\|\Pi_W(v)\| \leq \|v\|$ so Π_W is a bounded linear operator. $\square$

Riesz representation theorem

Theorem (Riesz representation theorem). If H is a Hilbert space, then for all $f \in H'$, there exists a unique $v \in H$ such that

$$f(u) = \langle u, v \rangle \text{ for all } u \in H.$$

Proof. For uniqueness, suppose $f(u) = \langle u, v \rangle = \langle u, \tilde{v} \rangle$. This implies that $\langle u, v - \tilde{v} \rangle = 0$ for all $u \in H$. Setting $u = v - \tilde{v}$ gives $v = \tilde{v}$.

Now we show existence. If $f = 0$, let $v = 0$. Suppose now $f \neq 0$. Then there exists $u_1 \in H$ such that $f(u_1) \neq 0$. Now let $u_0 = \frac{u_1}{f(u_1)}$, then $f(u_0) = 1$. Let

$$C = \{u \in H : f(u) = 1\} = f^{-1}(\{1\}).$$

Then C is a nonempty and closed subset of H . Claim that C is also convex. If $u_1, u_2 \in C$ and $t \in [0, 1]$, then

$$f(tu_1 + (1-t)u_2) = tf(u_1) + (1-t)f(u_2) = 1.$$

Therefore, C is also convex. This implies that there exists $v_0 \in C$ such that

$$v_0 = \inf_{u \in C} \|u\|.$$

Note that $v_0 \neq 0$ and let $v = \frac{v_0}{\|v_0\|^2}$. Claim this is the desired vector. Let $N = f^{-1}(\{0\})$. Then, $C = v_0 + N$ and $\|v_0\| = \inf_{w \in N} \|v_0 + w\|$. By a similar argument as a previous theorem, we have $v_0 \in N^\perp$. Let $u \in H$, then

$$f(u - f(u)v_0) = f(u) - f(u)f(v_0) = 0.$$

Therefore, $u - f(u)v_0 \in N$. Since $v_0 \in N^\perp$, we have $\langle u - f(u)v_0, v_0 \rangle = 0$. This implies that

$$\begin{aligned} \langle u, v_0 \rangle &= \frac{1}{\|v_0\|^2} \langle u, v_0 \rangle \\ &= \frac{1}{\|v_0\|^2} (\langle u - f(u)v_0, v_0 \rangle + f(u) \langle v_0, v_0 \rangle) \\ &= f(u), \end{aligned}$$

completing the proof. $\square$

2.5 Adjoint of a bounded linear operator on a Hilbert space

Theorem. Let H be a Hilbert space and $A : H \rightarrow H$ be a bounded linear operator. Then there exists a unique bounded linear operator $A^* : H \rightarrow H$ (the **adjoint** of A) such that for all $u, v \in H$,

$$\langle Au, Au \rangle = \langle u, u \rangle.$$

Moreover, $\|A^*\| = \|A\|$.

Proof. The uniqueness of A^* follows from a similar argument of uniqueness for Riesz representation theorem.

Let $v \in H$. Define $f_v : H \rightarrow \mathbb{C}$ by

$$f_v(u) = \langle Au, Au \rangle.$$

Then for any $u_1, u_2 \in H$ and $\lambda_1, \lambda_2 \in \mathbb{C}$, we have

$$\begin{aligned} f_v(\lambda_1 u_1 + \lambda_2 u_2) &= \langle A(\lambda_1 u_1 + \lambda_2 u_2), A(\lambda_1 u_1 + \lambda_2 u_2) \rangle \\ &= \lambda_1 \langle Au_1, Au_1 \rangle + \lambda_2 \langle Au_2, Au_2 \rangle \\ &= \lambda_1 f_v(u_1) + \lambda_2 f_v(u_2). \end{aligned}$$

Therefore, f_v is a linear map. Next, suppose $\|u\| = 1$, then

$$|f_v(u)| = |\langle Au, Au \rangle| \leq \|Au\| \|v\| \leq \|A\| \|v\|.$$

This shows that $f_v \in H'$. Then, by Riesz representation theorem, there is a unique $A^*v \in H$ such that for all $u \in H$,

$$f_v(u) = \langle Au, Au \rangle = \langle u, u \rangle.$$

Claim that $v \mapsto A^*v$ is linear. Let $v_1, v_2 \in H$ and $\lambda_1, \lambda_2 \in \mathbb{C}$, then for all $u \in H$, we have

$$\begin{aligned} \langle u, u \rangle &= \langle Au, Au \rangle \\ &= \overline{\lambda_1} \langle Au, Au \rangle + \overline{\lambda_2} \langle Au, Au \rangle \\ &= \overline{\lambda_1} \langle u, u \rangle + \overline{\lambda_2} \langle u, u \rangle \\ &= \langle u, u \rangle. \end{aligned}$$

This implies that

$$A^*(\lambda_1 v_1 + \lambda_2 v_2) = \lambda_1 A^*v_1 + \lambda_2 A^*v_2.$$

Therefore, A^* is a linear map. Next we show A^* is bounded and $\|A^*\| = \|A\|$. Suppose $\|v\| = 1$. If $A^*v = 0$, then clearly $\|A^*v\| \leq \|A\|$. Suppose now $A^*v \neq 0$, then

$$\|A^*v\|^2 = \langle A^*v, A^*v \rangle = \langle AA^*v, AA^*v \rangle \leq \|AA^*v\| \|v\| \leq \|A\| \|A^*v\|.$$

Therefore, $\|A^*\| \leq \|A\|$ and A^* is a bounded linear operator.

Note that for all $u, v \in H$,

$$\langle A^*u, A^*u \rangle = \overline{\langle v, v \rangle} = \overline{\langle Av, Av \rangle} = \langle u, u \rangle.$$

This implies that $A = (A^*)^*$. Therefore, $\|A\| = \|(A^*)^*\| \leq \|A^*\|$ and thus $\|A^*\| = \|A\|$. $\square$

Example. Suppose $u = (u_1, \dots, u_n) \in \mathbb{C}^n$ and $(Au)_i = \sum_j A_{ij}u_j$. Then,

$$\langle Au, Au \rangle = \sum_i (Au)_i \overline{(Au)_i} = \sum_{i,j} A_{ij}u_j \overline{(Au)_i} = \sum_j u_j \overline{\sum_i A_{ij}u_j} = \sum_j u_j \overline{A^* u}_j.$$

This implies that $(A^*)_{ij} = \overline{A_{ji}}$.

Example. Suppose $\{A_{ij}\}_{i,j=0}^\infty$ is a double sequence in $\mathbb{C}^n$ such that

$$\sum_{i,j} |A_{ij}|^2 = \lim_{N \rightarrow \infty} \sum_{i=1}^N \sum_{j=1}^N |A_{ij}|^2 < \infty.$$

Define $A : \ell^2 \rightarrow \ell^2$ by

$$Aa = \sum_{j=1}^\infty A_{ij}a_j,$$

where $a = \{a_j\}_{j=1}^\infty \in \ell^2$. Then, $A \in \mathcal{B}(\ell^2, \ell^2)$ and $(A^*)_{ij} = \overline{A_{ji}}$.

Example. Suppose $K \in C([0, 1] \times [0, 1])$ and define $A : L^2([0, 1]) \rightarrow L^2([0, 1])$ by

$$Af(x) = \int_0^1 K(x, y)f(y) dy.$$

Then we can verify that

$$A^*g(x) = \int_0^1 \overline{K(y, x)}g(y) dy.$$

Theorem. Suppose H is a Hilbert space and $A : H \rightarrow H$ is a bounded linear operator. Then,

$$(\text{range } A)^\perp = \ker(A^\perp).$$

Proof. $v \in \ker(A^*)$ if and only if $\langle u, v \rangle = 0$ for all $u \in H$. This is equivalent to $\langle Au, Au \rangle = 0$ for all $u \in H$, and this is equivalent to $v \in (\text{range } A)^\perp$. $\square$

Corollary. Suppose $\text{range } A$ is closed, then A is surjective if and only if A^* is injective.

With these useful tools we developed, we are soon going to discuss the solvability of linear equations, that is, equations in the form of $Au = v$. We take for granted that a bounded linear operator takes bounded sets to bounded sets in finite-dimensional spaces, and so we can find a convergent subsequence using Heine-Borel. So the point is that there is some compactness hidden in here in $\mathbb{R}^n$ and $\mathbb{C}^n$, so we need to study some facts about how compactness and Hilbert spaces before we can talk about solvability of equations.

Definition (Compact). If X is a metric space, we say $K \subset X$ is **compact** if every sequence in K has a subsequence converging to an element in K .

Example. Suppose H is an infinite dimensional separable Hilbert space, then $B[0, 1]$ is not compact.

Proof. Let $\{e_n\}_{n=1}^\infty$ be an orthonormal subset of H . Then for all $n \neq k$, we have

$$\|e_n - e_k\|^2 = \|e_n\|^2 + \|e_k\|^2 + 2 \operatorname{Re} \langle e_n, e_k \rangle = 2.$$

Therefore, $\{e_n\}_{n=1}^\infty$ cannot have a convergent subsequence. $\square$

Recall from Arzela-Ascoli, the extra condition for a set to be compact, other than closed and bounded, is **equicontinuity**. This motivates the following definition.

Definition (Equi-small tails). Let H be a Hilbert space. A subset $K \subset H$ has **equi-small tails** with respect to a countable orthonormal subset $\{e_k\}_{k=1}^\infty$ if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $v \in K$,

$$\sum_{k>N} |\langle v, e_k \rangle|^2 < \varepsilon^2.$$

Example. A finite set has equi-small tails with respect to any countable orthonormal subset $\{e_n\}_{n=1}^\infty$.

Theorem. Let H be a Hilbert space and $\{v_n\}_{n=1}^\infty$ a sequence with $v_n \rightarrow v$. Let $\{e_k\}_{k=1}^\infty$ be a countable orthonormal subset. Then,

1. $K = \{v_n : n \in \mathbb{N}\} \cup \{v\}$ is compact.
2. K has equi-small tails with respect to $\{e_k\}_{k=1}^\infty$.

Proof. 1. Exercise.

2. Since $v_n \rightarrow v$, there exists $M \in \mathbb{N}$ such that for all $n \geq M$, $\|v_n - v\| < \frac{\varepsilon}{2}$. Also, choose $N \in \mathbb{N}$ so large that

$$\sum_{k>N} |\langle v, e_k \rangle|^2 + \max_{1 \leq n \leq M-1} \sum_{k>N} |\langle v_n, e_k \rangle|^2 < \frac{\varepsilon^2}{4}.$$

Then,

$$\sum_{k>N} |\langle v, e_k \rangle|^2 < \frac{\varepsilon^2}{4} < \varepsilon^2.$$

and for all $1 \leq n \leq M-1$,

$$\sum_{k>N} |\langle v_n, e_k \rangle|^2 < \frac{\varepsilon^2}{4} < \varepsilon^2.$$

If $n \geq M$, by Bessel's inequality we have

$$\begin{aligned} \left[\sum_{k>N} |\langle v_n, v_n \rangle|^2 \right]^{\frac{1}{2}} &= \left[\sum_{k>N} |\langle v_n - v, v_n - v \rangle + \langle v, v \rangle|^2 \right]^{\frac{1}{2}} \\ &\leq \left[\sum_{k>N} |\langle v_n - v, v_n - v \rangle|^2 \right]^{\frac{1}{2}} + \left[\sum_{k>N} |\langle v, v \rangle|^2 \right]^{\frac{1}{2}} \\ &\leq \|v_n - v\| + \frac{\varepsilon}{2} \\ &< \varepsilon. \end{aligned}$$

This shows that K has equi-small tails with respect to $\{e_k\}_{k=1}^\infty$.

□

2.6 Compact subsets of a Hilbert space and finite-rank operators

Theorem. Let H be a separable Hilbert space, and $\{e_k\}_{k=1}^\infty$ be an orthonormal basis of H . Then $K \subset H$ is compact if and only if K is closed, bounded, and has equi-small tails with respect to $\{e_k\}_{k=1}^\infty$.

Proof. ($\implies$) Suppose K is compact, then K is closed and bounded. Suppose for contradiction that K does not have equi-small tails with respect to $\{e_k\}_{k=1}^\infty$. Then there exists $\varepsilon_0 > 0$ such that for all $N \in \mathbb{N}$, there exists $u_N \in K$ such that

$$\sum_{k>N} |\langle u_N, u_N \rangle|^2 \geq \varepsilon_0^2.$$

Since K is compact, there exists subsequence $\{v_n\}_{n=1}^\infty$ of $\{u_N\}_N$ such that $v_n \rightarrow v$ for some $v \in K$. Then for all $n \in \mathbb{N}$, we have

$$\sum_{k>n} |\langle v_n, v_n \rangle|^2 \geq \varepsilon_0^2.$$

Then $\{v_n : n \in \mathbb{N}\} \cup \{v\}$ does not have equi-small tails. This is a contradiction to the previous theorem. Therefore, K has equi-small tails with respect to $\{e_k\}_{k=1}^\infty$.

($\impliedby$) Suppose K is closed, bounded, and has equi-small tails. Let $\{u_n\}_{n=1}^\infty$ be a sequence in K . Since K is closed, it suffices to show that $\{u_n\}_{n=1}^\infty$ has a convergent subsequence. Since K is bounded, there exists $C \geq 0$ such that $\|u_n\| \leq C$. Then, for all $k, n \in \mathbb{N}$,

$$|\langle v_n, v_n \rangle| \leq \|u_n\| \|e_k\| \leq C.$$

That is, for any $k \in \mathbb{N}$, $\{\langle u_n, u_n \rangle\}_{n=1}^\infty$ is a bounded sequence in $\mathbb{C}$. This implies that there exists a subsequence $\{\langle u_{n_1(j)}, u_{n_1(j)} \rangle\}_{j=1}^\infty$ of $\{\langle u_n, u_n \rangle\}_{n=1}^\infty$ that converges in $\mathbb{C}$. Now $\{\langle u_{n_1(j)}, u_{n_1(j)} \rangle\}_{j=1}^\infty$

is a bounded sequence, there exists a subsequence $\{\langle u_{n_2(j)}, u_{n_2(j)} \rangle\}_{j=1}^\infty$ that converges. Note that $\lim_{j \rightarrow \infty} \langle u_{n_2(j)}, u_{n_2(j)} \rangle$ and $\lim_{j \rightarrow \infty} \langle u_{n_2(j)}, u_{n_2(j)} \rangle$ both exist. Inductively, for any $l \in \mathbb{N}$, there exists subsequence $\{n_l(j)\}_{j=1}^\infty$ of $\{n_{l-1}(j)\}_{j=1}^\infty$ such that for all $1 \leq k \leq l$, the limit $\lim_{j \rightarrow \infty} \langle u_{n_l(j)}, u_{n_l(j)} \rangle$ exists.

For all $l \in \mathbb{N}$, let $v_l = u_{n_l(l)}$, then $\{v_l\}_{l=1}^\infty$ is a subsequence of $\{u_n\}_{n=1}^\infty$ such that for any $k \in \mathbb{N}$, $\{\langle v_l, v_l \rangle\}_{l=1}^\infty$ converges. Claim that $\{v_l\}_{l=1}^\infty$ is Cauchy. This will complete the proof since the closed subset K of Hilbert space H is complete. Since K has equi-small tails, there exists $N \in \mathbb{N}$ such that for all $l \in \mathbb{N}$,

$$\sum_{k > N} |\langle v_l, v_l \rangle| < \frac{\varepsilon^2}{16}.$$

Since the N sequences $\{\langle v_l, v_l \rangle\}_{l=1}^\infty, \dots, \{\langle v_l, v_l \rangle\}_{l=1}^\infty$ all converges, there exists $M \in \mathbb{N}$ such that for all $l, m \geq M$, we have

$$\sum_{k=0}^N |\langle v_l, v_l \rangle - \langle v_m, v_m \rangle|^2 < \frac{\varepsilon^2}{4}.$$

Then for all $l, m \geq M$, we have

$$\begin{aligned} \|v_l - v_m\| &= \left[\sum_{k=0}^N |\langle v_l - v_m, v_l - v_m \rangle|^2 + \sum_{k > N} |\langle v_l - v_m, v_l - v_m \rangle|^2 \right]^{\frac{1}{2}} \\ &\leq \left[\sum_{k=0}^N |\langle v_l - v_m, v_l - v_m \rangle|^2 \right]^{\frac{1}{2}} + \left[\sum_{k > N} |\langle v_l - v_m, v_l - v_m \rangle|^2 \right]^{\frac{1}{2}} \\ &< \frac{\varepsilon}{2} + \left[\sum_{k > N} |\langle v_l, v_l \rangle|^2 \right]^{\frac{1}{2}} + \left[\sum_{k > N} |\langle v_m, v_m \rangle|^2 \right]^{\frac{1}{2}} \\ &< \varepsilon. \end{aligned}$$

□

Example. The Hilbert cube

$$K = \{ \{a_k\}_{k=1}^\infty \in \ell^2 : |a_k| \leq 2^{-k} \}$$

is compact.

Theorem. A subset K of a Hilbert space H is compact if and only if K is closed, bounded, and for all $\varepsilon > 0$, there exists a finite dimensional subspace $W \subset H$ such that for all $u \in K$,

$$\inf_{w \in W} \|u - w\| < \varepsilon.$$

Finite rank operators

Definition (Finite rank operator). A bounded linear operator $T \in \mathcal{B}(H)$ is a **finite rank operator** if $\text{range}(T)$, which is a subspace of H , is finite dimensional. Denote the set of all finite range operator on H with $\mathcal{R}(H)$.

Theorem. $\mathcal{R}(H)$ is a subspace of $\mathcal{B}(H)$.

Theorem. $T \in \mathcal{R}(H)$ if and only if there exists an orthonormal set $\{e_k\}_{k=0}^L$ and constants $\{c_{ij}\}_{i,j=0}^L \subset \mathbb{C}$ such that

$$Tu = \sum_{i,j=0}^L c_{ij} \langle u, e_j \rangle e_i.$$

Proof. ($\Leftarrow$). Clear.

($\Rightarrow$). Since $\text{range}(T)$ is finite dimensional, there exists orthonormal basis $\{\bar{e}_k\}_{k=0}^N$ such that

$$\begin{aligned} Tu &= \sum_{k=0}^N \langle Tu, \bar{e}_k \rangle \bar{e}_k \\ &= \sum_{k=0}^N \langle u, u \rangle \bar{e}_k \\ &= \sum_{k=0}^N \langle u, u \rangle \bar{e}_k. \end{aligned}$$

Let $\{e_0, \dots, e_L\}$ be the orthonormal subset of H obtained by applying the Gram-Schmidt process on $\{\bar{e}_0, \dots, \bar{e}_N, v_0, \dots, v_N\}$. Then, there exists a_{ki}, b_{kj} such that

$$\bar{e}_k = \sum_{i=0}^L a_{ki} e_i, \quad v_k = \sum_{j=0}^L b_{kj} e_j.$$

Then,

$$Tu = \sum_{k=0}^N \langle u, u \rangle \bar{e}_k = \sum_{i,j=0}^L \left(\sum_{k=0}^N a_{ki} \bar{b}_{kj} \right) \langle u, u \rangle e_i.$$

Setting $c_{ij} = \sum_{k=0}^N a_{ki} \bar{b}_{kj}$ completes the proof. $\square$

Theorem. The following holds:

1. $T \in \mathcal{R}(H)$ implies $T^* \in \mathcal{R}(H)$.
2. $T \in \mathcal{R}(H)$ and $A, B \in \mathcal{B}(H)$ implies $ATB \in \mathcal{R}(H)$.

Proof. 1. Write

$$Tu = \sum_{i,j=0}^L c_{ij} \langle w, w \rangle e_i.$$

Then, for any $u, v \in H$, we have

$$\begin{aligned} \langle u, u \rangle &= \langle Tu, Tu \rangle \\ &= \left\langle \sum_{i,j} c_{ij} \langle u, u \rangle e_i, \sum_{i,j} c_{ij} \langle u, u \rangle e_i \right\rangle \\ &= \sum_{i,j} c_{ij} \langle u, u \rangle \langle e_i, e_i \rangle \\ &= \langle u, u \rangle \\ &= \langle u, u \rangle. \end{aligned}$$

This implies that

$$T^*v = \sum_{i,j=0}^L \overline{c_{ij}} \langle v, v \rangle e_j,$$

so $T^* \in \mathcal{R}(H)$.

2. Exercise.

□

Now a good question to investigate is whether $\mathcal{R}(H)$ is a closed subspace in $\mathcal{B}(H)$? That is, if $T_n \in \mathcal{R}(H)$ and $\|T_n - T\| \rightarrow 0$, then is $T \in \mathcal{R}(H)$? The answer turns out to be no.

Example. Take $T_n : \ell^2 \rightarrow \ell^2$ defined by

$$T_n a = \left\{ \frac{a_1}{1}, \dots, \frac{a_n}{n}, 0, 0, \dots \right\}$$

for all $a = \{a_k\}_{k=1}^\infty \in \ell^2$. Then $T_n \in \mathcal{R}(H)$. Let $T : \ell^2 \rightarrow \ell^2$ defined by

$$Ta = \left\{ \frac{a_1}{1}, \frac{a_2}{2}, \frac{a_3}{3}, \frac{a_4}{4}, \dots \right\}.$$

We can verify that $\|T - T_n\| \leq \frac{1}{n+1}$ but T is not finite rank.

As $\mathcal{R}(H)$ is not closed, a question is what is the closure $\overline{\mathcal{R}(H)}$ in $\mathcal{B}(H)$. The answer to this is the set of **compact operators**.

Definition (Compact operator). $K \in \mathcal{B}(H)$ is a **compact operator** if $\overline{K(B[0,1])}$ is compact.

Theorem. $K \in \mathcal{B}(H)$ is a compact operator if and only if $K \in \overline{\mathcal{R}(H)}$.